

VentureScope AI - Product Requirements Document & Development Roadmap

Vision

Build an AI-powered Startup Intelligence Platform that analyzes startup ecosystems, predicts funding and growth outcomes, and provides explainable insights through analytics, machine learning, graph intelligence, and AI agents.

Problem Statement

Startup information is fragmented across YC, Product Hunt, GitHub, news sources, and company websites. Investors, founders, analysts, and students lack a unified intelligence platform for startup discovery and evaluation.

Target Users

Primary: Investors, VC Analysts, Founders. Secondary: Students, Researchers, Consultants.

MVP Features

Startup Explorer, Analytics Dashboard, Funding Prediction Engine, AI Startup Analyst, Startup Profiles, Search & Filtering.

Technology Stack

Frontend: Next.js, Tailwind, ShadCN, Framer Motion, Three.js. Backend: FastAPI. Database: PostgreSQL, DuckDB. Data Processing: Polars. ML: XGBoost, SHAP, Scikit-Learn. AI: Ollama + DeepSeek/Qwen.

Architecture

Data Sources -> Collection Layer -> PostgreSQL -> Analytics Engine / ML Models / AI Assistant -> FastAPI -> Next.js Frontend.

Database Tables

startups, founders, funding_events, github_metrics, news_mentions, predictions.

Jira Epics

1. Data Collection System. 2. Database Layer. 3. Analytics Engine. 4. ML Prediction System. 5. Frontend Dashboard. 6. AI Research Assistant.

Analytics Layer

Growth Score, Momentum Score, Founder Score, Funding Score, Investor Attractiveness Score.

Machine Learning Roadmap

Predict funding probability, shutdown probability, acquisition probability, and unicorn probability. Use SHAP for explainability.

Advanced Features

Startup Galaxy visualization, Startup Similarity Search, Emerging Sector Detection, Funding Timeline Forecast.

Execution Roadmap

Week 1: PRD, architecture, database schema. Week 2: Data collection pipelines. Week 3: Analytics engine. Week 4: ML models. Week 5: APIs. Week 6: Frontend. Week 7: AI assistant. Week 8+: Advanced features.